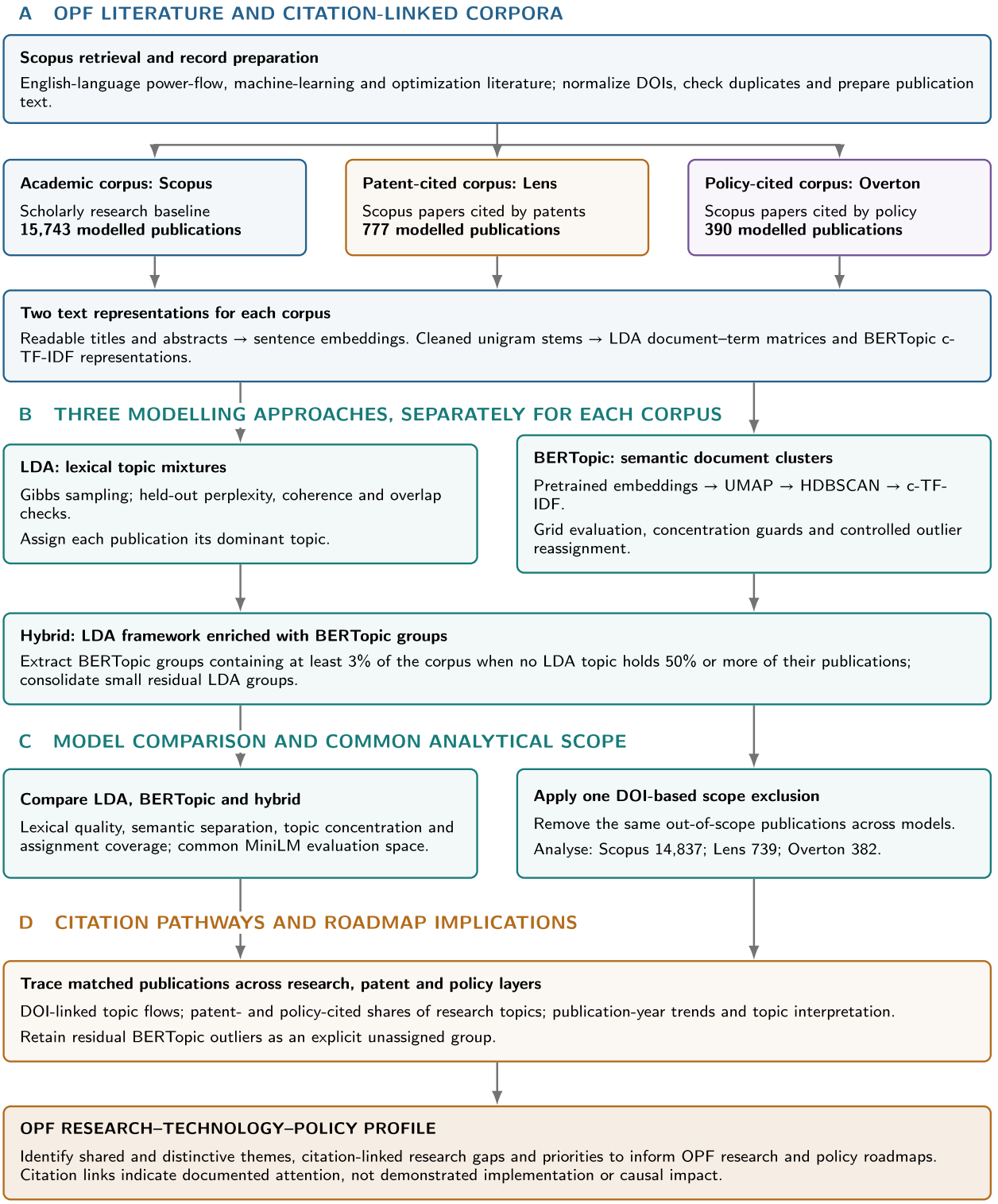

A OPF LITERATURE AND CITATION-LINKED CORPORA
Scopus retrieval and record preparation
English-language power-flow, machine-learning and optimization literature; normalize DOIs, check duplicates and prepare publication text.
Academic corpus: Scopus
Scholarly research baseline
15,743 modelled publications
Patent-cited corpus: Lens
Scopus papers cited by patents
777 modelled publications
Policy-cited corpus: Overton
Scopus papers cited by policy
390 modelled publications
Two text representations for each corpus
Readable titles and abstracts → sentence embeddings. Cleaned unigram stems → LDA document–term matrices and BERTopic c-TF-IDF representations.
B THREE MODELLING APPROACHES, SEPARATELY FOR EACH CORPUS
LDA: lexical topic mixtures
Gibbs sampling; held-out perplexity, coherence and overlap checks.
Assign each publication its dominant topic.
BERTopic: semantic document clusters
Pretrained embeddings → UMAP → HDBSCAN → c-TF-IDF.
Grid evaluation, concentration guards and controlled outlier reassignment.
Hybrid: LDA framework enriched with BERTopic groups
Extract BERTopic groups containing at least 3% of the corpus when no LDA topic holds 50% or more of their publications; consolidate small residual LDA groups.
C MODEL COMPARISON AND COMMON ANALYTICAL SCOPE
Compare LDA, BERTopic and hybrid
Lexical quality, semantic separation, topic concentration and assignment coverage; common MiniLM evaluation space.
Apply one DOI-based scope exclusion
Remove the same out-of-scope publications across models.
Analyse: Scopus 14,837; Lens 739; Overton 382.
D CITATION PATHWAYS AND ROADMAP IMPLICATIONS
Trace matched publications across research, patent and policy layers
DOI-linked topic flows; patent- and policy-cited shares of research topics; publication-year trends and topic interpretation.
Retain residual BERTopic outliers as an explicit unassigned group.
OPF RESEARCH–TECHNOLOGY–POLICY PROFILE
Identify shared and distinctive themes, citation-linked research gaps and priorities to inform OPF research and policy roadmaps. Citation links indicate documented attention, not demonstrated implementation or causal impact.